

Project Overview Report

AI Resume Analyzer and Job Recommendation System

Introduction

This project is designed as an educational, practical tool that reads a candidate's resume (in PDF or DOCX format), extracts their core technical skills, compares them against real-world job role requirements using Natural Language Processing (NLP), and provides helpful feedback—including a custom learning roadmap powered by Groq AI.

How the System Works (Project Workflow)

Building this application required breaking down the problem into sequential, manageable steps. Here is how data moves through the application:

- **Step 1: Resume Upload**
- **Step 2: Text Extraction & Cleaning**
- **Step 3: Skill Detection**
- **Step 4: Matching & Scoring**
- **Step 5: Skill Gap & AI Roadmap**

Technical Tools and Tech Stack

Every part of this project was built using standard, industry-proven Python libraries running inside Visual Studio Code (VS Code):

- **Programming Language:** Python 3.10
- **Document Parsing:** pypdf and python-docx
- **Text Vectorization:** scikit-learn (TF-IDF)
- **User Interface:** Streamlit
- **AI Feedback:** Groq API (Llama 3.1)
- **Data Handling:** Pandas for managing job roles, required skills, and matching scores.

Key Features Implemented

The final application successfully achieves all baseline project objectives outlined in my curriculum guide:

- **Flexible Uploads:** Accepts both PDF and DOCX files
- **Accurate Keyword Matching:** Identifies over 25+ technical skills ranging from Python and SQL to Deep Learning and Docker.
- **Multi-Role Comparison:** Evaluates resumes across 5 key roles including Data Analyst, Machine Learning Engineer, AI Engineer, NLP Engineer, and Computer Vision Engineer.
- **Interactive Dashboard:** Displays match percentages, detected skills, and role rankings instantly.
- **AI-Powered Guidance:** Uses the Groq LLM to provide structured, weekly study plans for any skill gaps identified.